

CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS

FAMILY TREE

Aim:

To implement a family tree using Prolog and derive relationships between family members.

Objective:

To represent parent-child relationships.

To derive father, mother, sibling and grandparent relationships. To demonstrate rule-based reasoning.

Algorithm: Start.

Define parent facts.

Define male and female facts.

Define rules for father, mother, sibling and grandparent. Query the required relationship.

Display the result. Stop.

Flowchart:

START

| v

Define Family Facts

| v

Define Relationship Rules

| v

Enter Query

| v

Match Facts and Rules

| v

Display Relationship

| v

STOP

Prolog Program:

% Family Tree

male(ramesh). male(arun). male(kiran).

female(sita). female(priya). female(neha).

parent(ramesh, arun). parent(sita, arun).

parent(ramesh, priya). parent(sita, priya).

parent(arun, kiran). parent(priya, neha).

father(X, Y) :-male(X),

parent(X, Y).

mother(X, Y) :-female(X), parent(X, Y).

grandparent(X, Z) :-parent(X, Y),

parent(Y, Z).

sibling(X, Y) :-

parent(P, X),

parent(P, Y), X \= Y..

Output:

Result:

Thus, the family tree is successfully implemented using Prolog facts and rules.